

1 Experiments

We evaluate our pipeline on the SoccerNet Game State Reconstruction (GSR) dataset and compare against its official deep-learning baseline. Our experiments are organized around three questions: is a classical homography *accurate enough* for minimap localization (Experiment 1), does each component of the pipeline *earn its place* (Experiment 2), and is the method *cheap enough* to deliver on its accessibility motivation (Experiment 3). This draft covers the setup and Experiment 1.

1.1 Experimental Setup

Dataset. We use the validation split of the SoccerNet GSR dataset: 58 broadcast clips, each 30 seconds long at 25 frames per second (750 frames of 1920×1080 video), spanning varied stadiums, lighting, and camera work. Every frame is annotated with pitch-line points and with player, goalkeeper, referee, and ball bounding boxes, each carrying a ground-truth pitch coordinate.

Baseline. We compare against SoccerNet’s `sn-gamestate` baseline, a deep multi-stage pipeline whose camera calibration is a trained neural network producing a full 3D camera model. We obtain its predictions from the official released baseline tracker state, which provides, for every clip, the baseline’s detected pitch lines, detected boxes, and predicted pitch positions. We note that the baseline does not place the ball on its 2D minimap at all: its radar visualization explicitly skips ball detections, and its detector in any case rarely localizes the ball (e.g. producing ball detections in only 0 and 5 of 750 frames on two inspected clips). In contrast, our pipeline reconstructs the ball explicitly, including a physics-based correction for airborne frames.

Inputs and metric. Our pipeline consumes only per-frame pitch lines and player/ball detections; obtaining these is a standard perception problem outside our scope, so we evaluate our geometry under two input regimes — the dataset’s ground-truth annotations, and the baseline’s own (imperfect) detected lines and boxes. The latter lets us compare geometry methods on identical inputs. Unless stated otherwise, error is the Euclidean distance, in metres on the pitch, between a predicted position and its ground-truth pitch coordinate.

1.2 Experiment 1: Player Localization Accuracy

Goal. Determine whether a classical planar homography is accurate enough for minimap localization, and how it compares to the baseline’s learned 3D calibration.

Protocol. We measure per-detection localization error for all on-pitch persons across all 58 validation clips (the ball is excluded here, as its ground-truth pitch coordinate is unreliable when airborne; the ball is addressed in Experiment 2), under three conditions:

- **Ours (GT lines)** — our homography estimated from the dataset’s ground-truth pitch lines, projecting ground-truth foot points. With perfect inputs, this is an upper bound on our geometry.
- **Ours (SoccerNet inputs)** — our homography estimated from the baseline’s *detected* pitch lines, projecting the baseline’s *detected* foot points.
- **SoccerNet** — the baseline’s own predicted positions from its 3D calibration.

The latter two conditions operate on identical detections, so any difference between them isolates the geometry method. To score the baseline’s detections against ground truth, each detection is matched to a ground-truth box by greedy image-space IoU (threshold 0.5); unmatched detections are a perception-recall issue and are excluded.

Table 1: Player localization error against ground truth, over all 58 validation clips. “Ours (GT lines)” uses perfect inputs and is an upper bound; the other two conditions share SoccerNet’s detections and thus isolate the geometry method. Median and p90 are in metres; the last three columns are the percentage of detections within the given radius. Best per column in bold.

Condition	Median (m)	p90 (m)	<1 m (%)	<2 m (%)	<5 m (%)
Ours (GT lines)	0.48	1.78	78.1	91.2	96.8
Ours (SoccerNet inputs)	0.84	3.60	57.6	80.3	93.0
SoccerNet	0.98	20.22	50.7	69.7	78.9

Results. Table ?? reports the error distribution, aggregated over 696,776 detections for the ground-truth-input condition and 569,075 matched detections for the two detected-input conditions.

With ground-truth lines our homography localizes players to a median of 0.48 m, with 91% of detections within 2 m. Fed the baseline’s own detected lines and boxes — a controlled, like-for-like test of geometry — our method achieves a median error of 0.84 m, *lower* than the baseline’s 0.98 m on the identical detections, and wins at every tolerance (e.g. 80.3% vs. 69.7% within 2 m). The gap is most pronounced in the tail: our 90th-percentile error is 3.60 m, against 20.22 m for the baseline — one in ten of the baseline’s predictions is off by more than 20 m. The mean error is the lone metric favouring the baseline (7.90 m vs. our 8.70 m); this statistic is dominated by a thin tail of extreme frames — our method in fact produces *fewer* large errors (7% beyond 5 m vs. the baseline’s 21%) — so we report the robust median and threshold fractions as the primary measures.

Figure ?? shows a representative frame. Our reconstruction, under both input regimes, reproduces the ground-truth player configuration, whereas the baseline’s calibration shifts the entire formation by roughly 15 m. The 0.48 $\rightarrow$ 0.84 m increase between our two input regimes quantifies the cost of imperfect detected lines.

Takeaway. A classical planar homography matches and, on this evidence, exceeds a trained 3D-calibration network for minimap player localization, while requiring no training, no learned weights, and no GPU.

1.3 Experiment 2: Component Ablation

Goal. Our pipeline adds three components on top of a plain homography fit: circle-based handling of degenerate camera views, temporal smoothing, and physics-based gating of the airborne ball. We disable each in turn and measure the metric it targets, over all 58 clips (43,500 frames), to confirm each earns its place. Results are summarized in Table ??.

Circle-degeneracy handling. Many broadcast frames show only mutually parallel pitch lines — for example the two sidelines and the halfway line — which cannot constrain an eight-degree-of-freedom homography. For these frames our method adds point correspondences derived from the central circle. With this component disabled, the fraction of frames yielding a valid homography falls from 96.3% to 81.6%: circle handling recovers 14.7% of all frames (roughly 6,400 frames) that would otherwise have no minimap at all.

Temporal smoothing. Independently fitted homographies jitter from frame to frame. We measure the per-frame displacement of fixed probe points projected onto the pitch; as this quantity is heavy-tailed (a failed fit sends probe points near-infinitely far), we report the median and tail percentiles. The median-filter smoothing reduces the 90th-percentile displacement from 3.90 m to 2.89 m and the 95th from 12.79 m to 9.06 m (−26% / −29%), and the rate of large (>5 m) frame-to-frame jumps from 8.7% to 7.2%, while leaving the median essentially unchanged (0.37 m $\rightarrow$ 0.31 m). It thus suppresses jitter without lagging genuine camera motion.

Airborne-ball gating. A ground-plane homography cannot localize an airborne ball: its image point can back-project far off the pitch. Without gating, the raw ball projection lands off the pitch in 3,895

Table 2: Component ablation over all 58 validation clips (43,500 frames). Each component is disabled in turn and evaluated by the metric it targets. “Full” is the complete pipeline; “Ablated” removes that one component.

Component ablated	Metric	Full	Ablated
Circle-degeneracy handling	Frames with valid homography (%)	96.3	81.6
Temporal smoothing	p95 probe displacement (m)	9.06	12.79
Airborne-ball gating	Off-pitch ball frames	0	3895

frames; our physics-based gate and trajectory interpolation reduce this to zero. On frames where the ball’s ground-truth position is itself reliable, gating leaves accuracy unchanged (median error $0.63 \rightarrow 0.64$ m, mean $1.68 \rightarrow 1.59$ m) — it eliminates the catastrophic frames at no cost to the rest.

Takeaway. All three components yield measurable improvements: circle handling and ball gating eliminate outright failures (unsolvable frames and off-pitch ball projections), while temporal smoothing reduces residual jitter without sacrificing responsiveness to real camera motion.

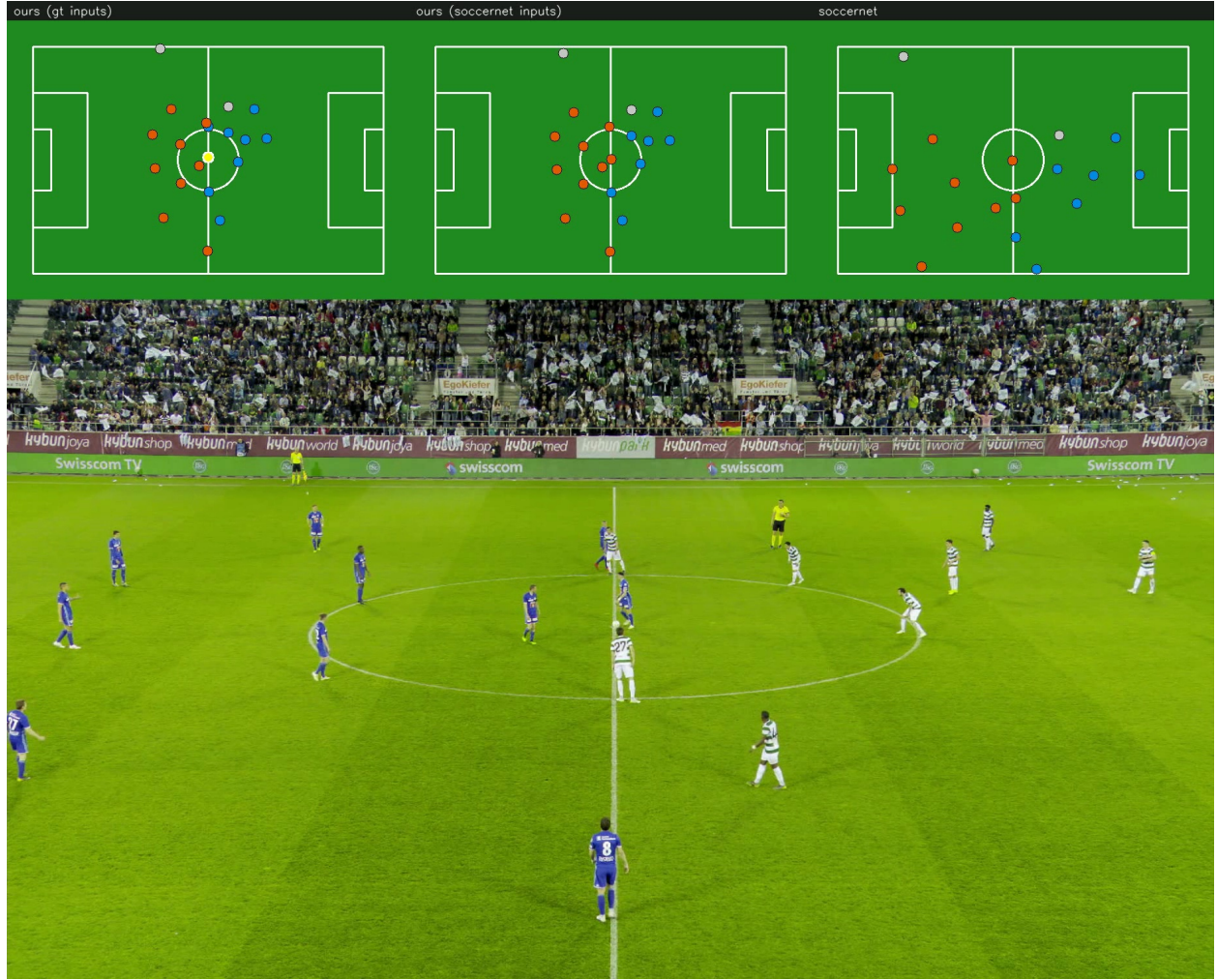


Figure 1: Localization on clip SNGS-095 (an illustrative example; the full distribution is in Table ??). Top, left to right: minimaps from our homography on ground-truth inputs, our homography on SoccerNet’s detected inputs, and SoccerNet’s own 3D calibration. Bottom: the source broadcast frame. Both of our reconstructions match the true formation; the baseline displaces it by ~ 15 m. The ball (yellow) appears only in our ground-truth-input panel: our pipeline reconstructs it, whereas the SoccerNet baseline neither reliably detects the ball nor places it on the minimap.